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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	404 Sleep Not Found
Project / Solution Name	StockSync AI for Orbyte
Name (Leader)	Ling Xin Qi
Email & Phone (Leader)	xinqi052@gmail.com & 0188704593
University Affiliation(s)	Universiti Sains Malaysia
Live Prototype URL	https://stock-sync-ai.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/Lxq1001/StockSync-AI

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

StockSync AI is a web-based inventory management prototype built with React and TypeScript. Vite supports frontend development and building, while Tailwind CSS and Recharts provide styling and data visualisation. The React frontend communicates with a Node.js/Express backend API, which loads inventory, supplier, sales transaction, and import data from local JSON files and serves the combined dataset through the `/api/stock-dataset` endpoint. Zod validates the dataset structure before it is used by the application.

For restocking recommendations, inventory data can be sent to a Python/FastAPI analysis service through the `/api/analyze` endpoint. The service uses `handler.calculate_restock()` to identify items with less than seven days of estimated stock cover and recommend suitable reorder quantities. Users can review these suggestions, accept the recommended quantities, or manually specify quantities for other items they wish to reorder.

When an order is initiated, the system prepares a supplier email containing the selected items, order quantities, and calculated prices. Users can review and edit the email before sending it. The application also supports monitoring stock levels, adjusting inventory quantities and prices, and managing order statuses. The prototype uses local JSON files as its primary data source rather than a database.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.x	Main application programming language	Apache-2.0
Python	Implements the restocking analysis service	PSF License

Name & Version	Purpose in Project	License
React	Builds the frontend user interface	MIT
Vite	Frontend development and build tooling	MIT
Tailwind CSS	Styles the user interface	MIT
Node.js	Runs the JavaScript backend	MIT
Express 5.x	Provides the backend API	MIT
FastAPI	Provides the Python analysis API	MIT
Zod	Validates dataset structures	MIT
Recharts 2.15.x	Displays charts and data visualisations	MIT
Pydantic	Validates incoming data for the Python API	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
None	The application does not directly call external third-party APIs. Supplier emails are handled by redirecting users to Gmail to review and send the email.	N/A

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Cloud-based development environment for coding, running, and testing StockSync AI.	Core / Malaysia

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None	N/A	N/A	N/A

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Inventory Dataset (mock_db.json)	Created by the project team	Synthetic	N/A	NO
Supplier Dataset (suppliers.json)	Created by the project team	Synthetic	N/A	NO
Sales Transaction Dataset (sales_transaction.json)	Created by the project team	Synthetic	N/A	NO
Import Dataset (import_data.json)	Created by the project team	Synthetic	N/A	NO

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT (OpenAI)	Coding assistance, debugging, problem-solving and technical explanations	Moderate
Gemini (Google)	Coding assistance, debugging, problem-solving and technical explanations	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

We used AI tools as supplementary resources to support our development process and understanding of technical concepts. All AI-generated suggestions were critically evaluated against our project requirements before being considered for implementation. We independently reviewed and tested the relevant code, debugged errors, and modified the output where necessary to ensure it functioned as intended. Suggestions that were inaccurate or unsuitable were rejected or revised. Our team made the final technical decisions and took full responsibility for the accuracy, quality, security, and functionality of StockSync AI. AI assistance complemented our development efforts rather than replacing our own judgement and problem-solving.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit — cloud-based development platform	Provided a cloud-based environment for developing and testing the StockSync AI application.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provided a valuable cloud-based environment that supported our development, testing, and implementation of StockSync AI. It helped streamline our development workflow and facilitated the testing of our application's features. Although alternative development environments could also support the project, Replit offered a convenient platform for our team to develop and refine the prototype. The core inventory management and restocking recommendation functionalities are implemented through our own application code.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

StockSync AI uses inventory, supplier, sales transaction, and import data to support stock monitoring and restocking recommendations. The prototype primarily uses local JSON files for its datasets. It does not directly integrate with external AI APIs. When users prepare supplier orders, the system may redirect them to Gmail to review and send the order email. Any information included in the email is subject to Gmail's data-handling practices. The prototype's data retention depends on the storage method used, and local JSON data remains available until the files are modified or deleted. We take appropriate measures to protect project data and prevent the exposure of sensitive information or credentials during development and demonstration.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

StockSync AI has several technical limitations and potential risks. Our restocking recommendations depend on the accuracy, completeness, and timeliness of inventory and sales data. Inaccurate or outdated information may lead to unsuitable reorder quantities. Unexpected changes in customer demand, supplier delays, and stock availability may also affect recommendation accuracy. Although our system reduces manual workload by identifying low-stock items, recommending reorder quantities, and preparing supplier orders, it cannot automatically track supplier email replies or detect negotiations between managers and suppliers. Consequently, purchase order statuses cannot be updated automatically based on email responses or negotiation outcomes, requiring manual status updates. Our prototype also relies on local JSON files, which may limit data persistence, concurrent access, and scalability compared with a production database. Further testing, email integration and improved data management could enhance automation, reliability, and future scalability.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☑ Any use of official SDC sponsor resources has been declared in Section 5.

☑ The prototype described here is accessible to judges through the links provided in Section 1.

☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



22nd October 2026